

**17ce049**

# Practical 1

Aim-Build a regression model and predict the per capita income fo canadian citizens in year 2020

In [ ]:

```
import pandas as pd
import numpy as np
from sklearn import linear_model
import matplotlib.pyplot as plt
```

In [7]:

```
df = pd.read_csv('canada_per_capita_income.csv')
df.head()
```

Out[7]:

|   | year | income      |
|---|------|-------------|
| 0 | 1970 | 3399.299037 |
| 1 | 1971 | 3768.297935 |
| 2 | 1972 | 4251.175484 |
| 3 | 1973 | 4804.463248 |
| 4 | 1974 | 5576.514583 |

In [9]:

```
X=df.year
y=df.income
```

In [10]:

```
model=linear_model.LinearRegression()
model.fit(np.array(X).reshape(-1,1), y)
```

Out[10]:

```
LinearRegression(copy_X=True, fit_intercept=True, n_jobs=None, normalize=False)
```

In [11]:

```
print(f'Income per capita for year 2020 is{model.predict([[2020]])[0]}')
```

Income per capita for year 2020 is41288.69409441762

In [12]:

```
print(f"intercept is{model.intercept_}and coefficient is{model.coef_[0]}")
```

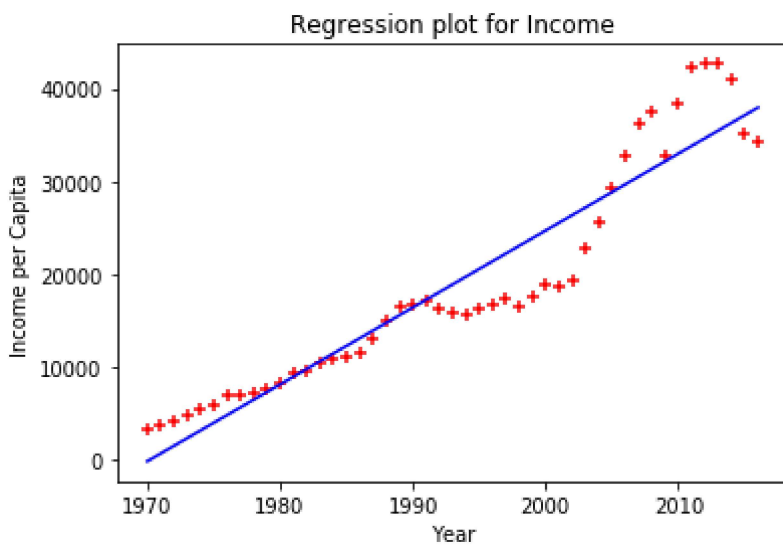
intercept is-1632210.7578554575and coefficient is828.4650752227104

In [16]:

```
%matplotlib inline
plt.xlabel('Year')
plt.ylabel('Income per Capita')
plt.scatter(X,y,color='red',marker='+')
plt.plot(X,model.predict(np.array(X).reshape(-1,1)),color='blue')
plt.title('Regression plot for Income')
```

Out[16]:

Text(0.5, 1.0, 'Regression plot for Income')



In [ ]: